

Sequential File Allocation

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int disk[50] = {0};
    int startBlock, len, i, j, flag;
    char choice;
    printf("--- Sequential File Allocation ---\n");
    do {
        flag = 0;
        printf("\nEnter starting block (0-49): ");
        scanf("%d", &startBlock);
        printf("Enter length of file: ");
        scanf("%d", &len);
        if (startBlock + len > 50) {
            printf("Error: File exceeds disk space.\n");
        } else {
            for (i = startBlock; i < startBlock + len; i++) {
                if (disk[i] == 1) {
                    flag = 1; // Block already allocated
                    break;
                }
            }
            if (flag == 1) {
                printf("Error: Contiguous blocks are not free.\n");
            } else {
                for (j = startBlock; j < startBlock + len; j++) {
                    disk[j] = 1;
                }
                printf("File allocated successfully from block %d to %d.\n", startBlock, startBlock +
len - 1);
            }
        }
        printf("Do you want to allocate another file? (y/n): ");
        scanf(" %c", &choice);
    } while (choice == 'y' || choice == 'Y');

    return 0;
}
```

```

1WA24CS218
--- Sequential File Allocation ---

Enter starting block (0-49): 4
Enter length of file: 6
File allocated successfully from block 4 to 9.
Do you want to allocate another file? (y/n): y

Enter starting block (0-49): 5
Enter length of file: 3
Error: Contiguous blocks are not free.
Do you want to allocate another file? (y/n): n

...Program finished with exit code 0
Press ENTER to exit console.

```

Indexed File Allocation

```

#include <stdio.h>
int main() {
    int disk[50] = {0};
    int indexBlock, n, i, flag;
    int blocks[50];
    char choice;
    printf("--- Indexed File Allocation ---\n");

    do {
        printf("\nEnter index block (0-49): ");
        scanf("%d", &indexBlock);
        if (disk[indexBlock] == 1) {
            printf("Error: Index block %d is already allocated.\n", indexBlock);
        } else {
            printf("Enter number of blocks for the file: ");
            scanf("%d", &n);
            printf("Enter the %d block numbers: ", n);
            flag = 0;

```

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    for (i = 0; i < n; i++) {
        scanf("%d", &blocks[i]);
        if (disk[blocks[i]] == 1) {
            flag = 1;
        }
    }

    if (flag == 1) {
        printf("Error: One or more blocks are already allocated. Allocation failed.\n");
    } else {
        disk[indexBlock] = 1;
        for (i = 0; i < n; i++) {
            disk[blocks[i]] = 1;
        }

        printf("File allocated successfully.\n");
        printf("Index Block: %d\n", indexBlock);
        printf("Allocated Data Blocks: ");

        for (i = 0; i < n; i++) {
            printf("%d ", blocks[i]);
        }
        printf("\n");
    }
}

printf("Do you want to allocate another file? (y/n): ");
scanf(" %c", &choice);
}

while (choice == 'y' || choice == 'Y');
return 0;
}

```

```

1WA24CS218
--- Indexed File Allocation ---

Enter index block (0-49): 8
Enter number of blocks for the file: 4
Enter the 4 block numbers: 12 3 45 31
File allocated successfully.
Index Block: 8
Allocated Data Blocks: 12 3 45 31
Do you want to allocate another file? (y/n): y

Enter index block (0-49): 12
Error: Index block 12 is already allocated.
Do you want to allocate another file? (y/n): n

...Program finished with exit code 0
Press ENTER to exit console.

```

Linked File Allocation

```

#include <stdio.h>
#include <stdlib.h>
struct Block {
    int allocated;
    int next;
};
int main() {
    struct Block disk[50];
    int startBlock, n, i, currentBlock, nextBlock;
    char choice;
    for (i = 0; i < 50; i++) {
        disk[i].allocated = 0;
        disk[i].next = -1;
    }
    printf("--- Linked File Allocation ---\n");
    do {
        printf("\nEnter starting block (0-49): ");
        scanf("%d", &startBlock);
        if (disk[startBlock].allocated == 1) {

```

```

    printf("Error: Starting block is already allocated.\n");
} else {
    printf("Enter number of blocks required: ");
    scanf("%d", &n);
    disk[startBlock].allocated = 1;
    currentBlock = startBlock;
    if(n == 1) {
        printf("File allocated successfully. Chain: %d -> NULL\n", startBlock);
    }
    else {
        printf("Enter the remaining %d blocks to form the chain:\n", n - 1);
        int success = 1;
        for (i = 1; i < n; i++) {
            printf("Next block after %d: ", currentBlock);
            scanf("%d", &nextBlock);
            if (disk[nextBlock].allocated == 1) {
                printf("Error: Block %d is already allocated! Aborting chain.\n", nextBlock);
                success = 0;
                break;
            } else {
                disk[nextBlock].allocated = 1;
                disk[currentBlock].next = nextBlock;
                currentBlock = nextBlock;
            }
        }
        if (success) {
            printf("File allocated successfully. Chain: ");
            int temp = startBlock;
            while (temp != -1) {
                printf("%d -> ", temp);
                temp = disk[temp].next;
            }
            printf("NULL\n");
        }
    }
}
printf("Do you want to allocate another file? (y/n): ");
scanf(" %c", &choice);
}

```

```
while (choice == 'y' || choice == 'Y');  
    return 0;  
}
```

```
1WA24CS218  
--- Linked File Allocation ---  
  
Enter starting block (0-49): 3  
Enter number of blocks required: 4  
Enter the remaining 3 blocks to form the chain:  
Next block after 3: 15  
Next block after 15: 37  
Next block after 37: 21  
File allocated successfully. Chain: 3 -> 15 -> 37 -> 21 -> NULL  
Do you want to allocate another file? (y/n): n  
  
...Program finished with exit code 0  
Press ENTER to exit console.█
```